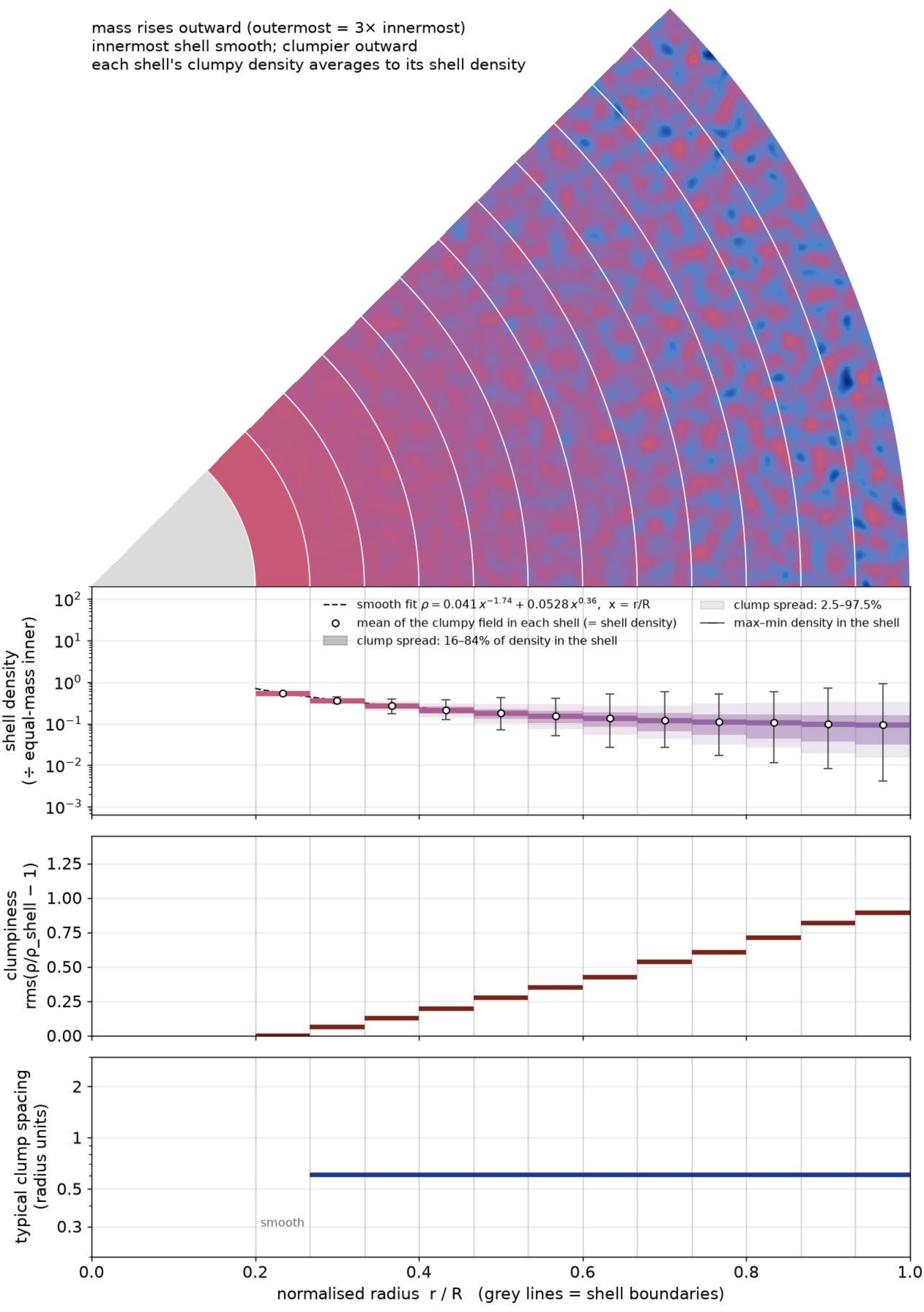
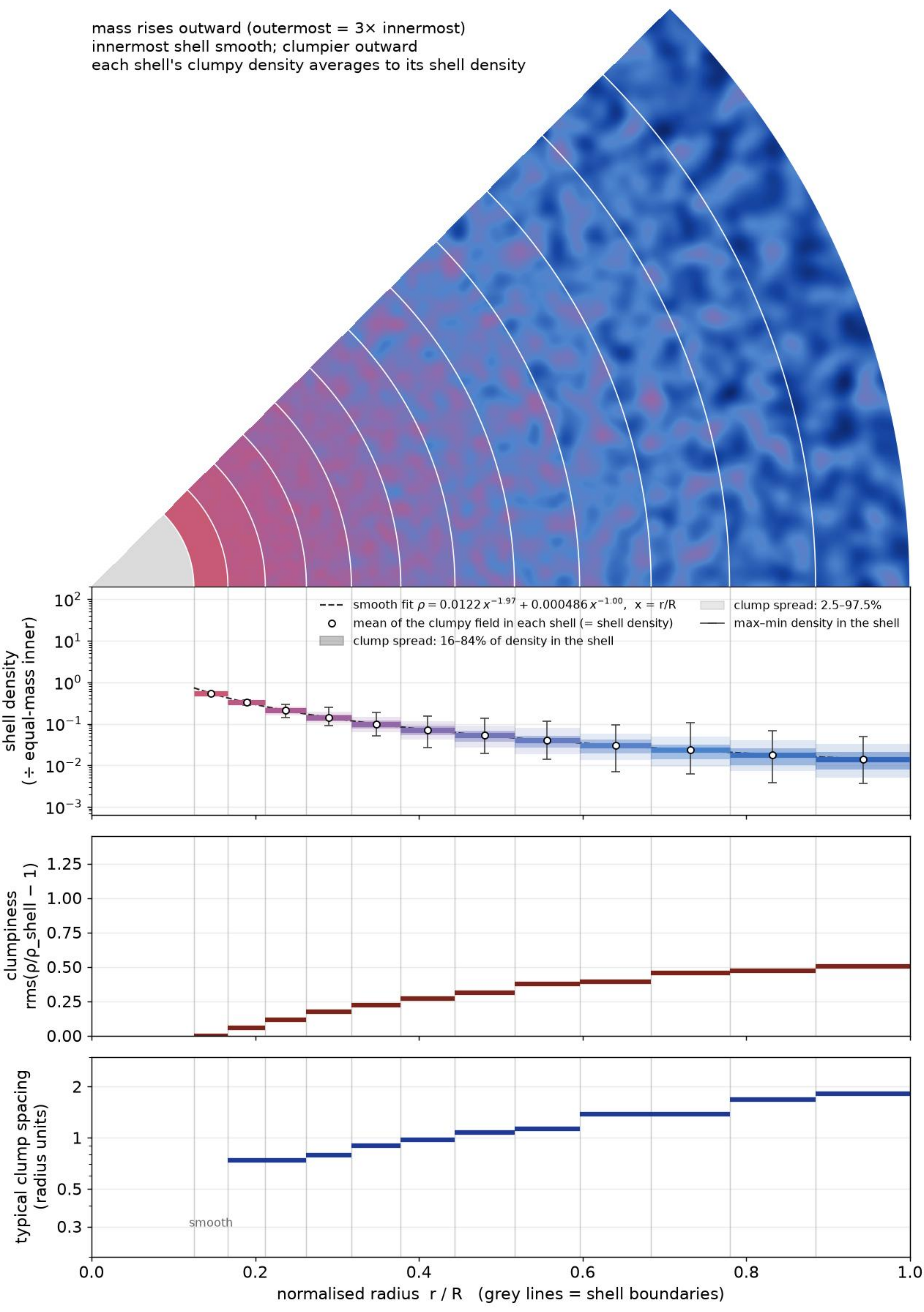
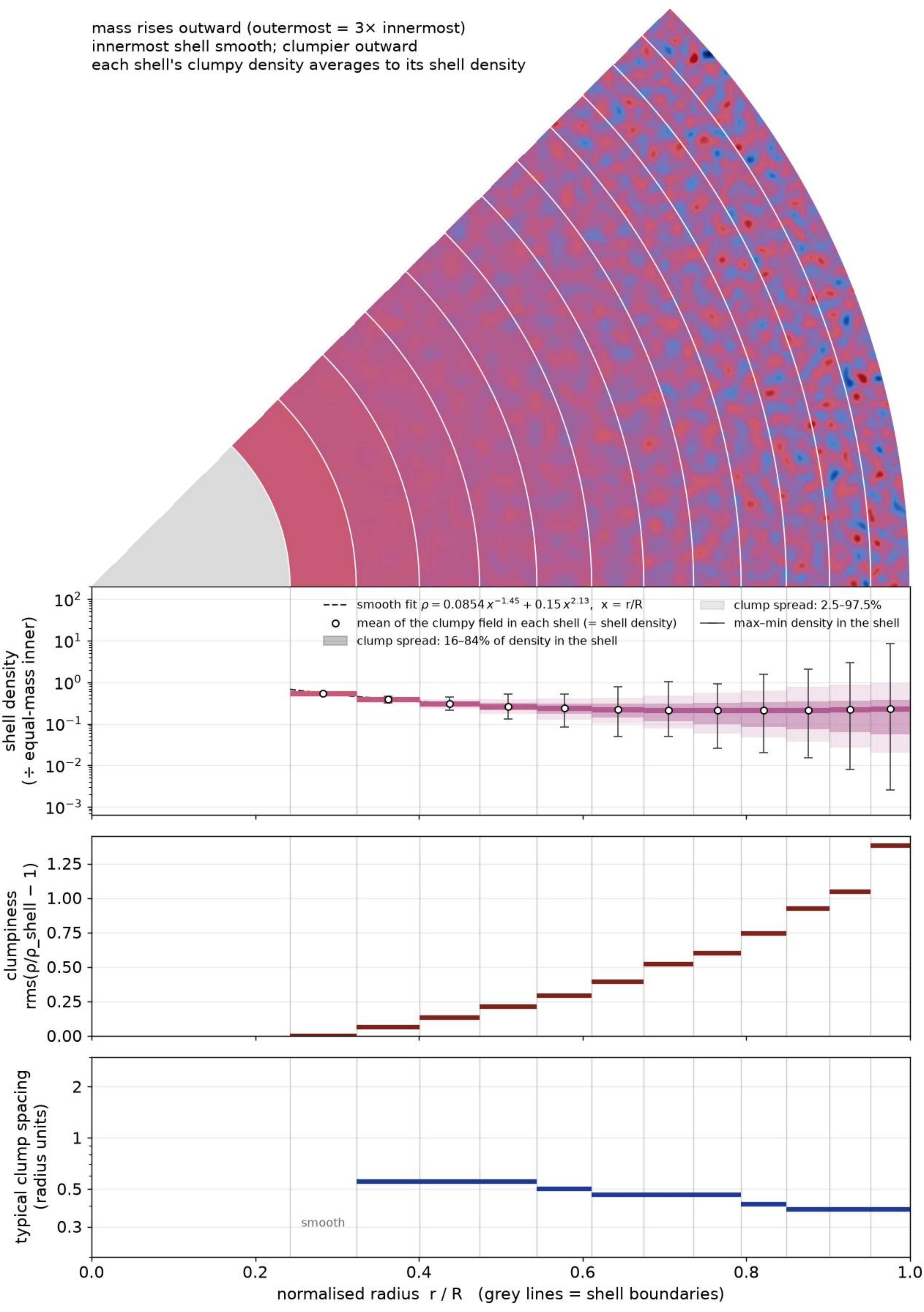
Structure in outer-heavy shells — Constant shell spacing

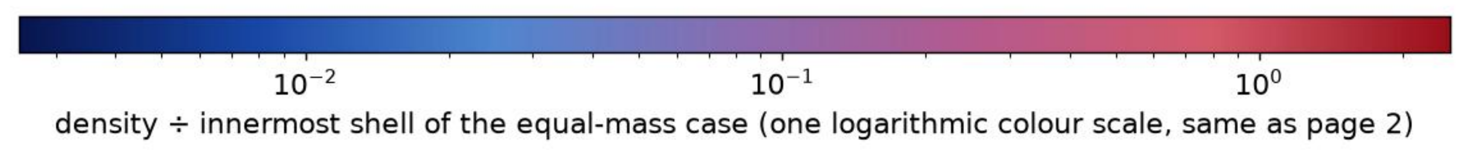


Structure in outer-heavy shells — Widening shell spacing (geometric growth)



Structure in outer-heavy shells — Narrowing shell spacing (geometric shrink)


mass rises outward (outermost = 3× innermost)
innermost shell smooth; clumpier outward
each shell's clumpy density averages to its shell density
smooth fit $\rho = 0.0854\,x^{-1.45} + 0.15\,x^{2.13}$, x = r/R
mean of the clumpy field in each shell (= shell density)
clump spread: 16–84% of density in the shell
clump spread: 2.5–97.5%
max–min density in the shell
shell density (÷ equal-mass inner)
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
clumpiness rms(ρ/ρ_shell − 1)
1.25
1.00
0.75
0.50
0.25
0.00
typical clump spacing (radius units)
2
1
0.5
0.3
smooth
0.0
0.2
0.4
0.6
0.8
1.0
normalised radius r / R (grey lines = shell boundaries)


10^{-2}
10^{-1}
10^0
density ÷ innermost shell of the equal-mass case (one logarithmic colour scale, same as page 2)